Artificial Intelligence based Models to Support Water Quality Prediction using Machine Learning Approach

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***Abstract*— The purpose of this study is to simulate the propagation of bio-contamination risk in Water Distribution System (WDS). This will be accomplished by putting an emphasis on source identification and response modelling. This will be accomplished in the WDS via the process of simulating the bio-contamination risk propagation under real environmental circumstances. This will be done in order to accomplish this goal. Due to the failure to take into consideration the ability for the bulk of the pollutants to undergo chemical degradation and the neglecting of dynamic responses, an overestimation of the population that is exposed has resulted. The study presented here takes into account the connected change in a variety of water quality indicators that are caused by the mixing of the pollutants. These measurements include the total organic carbon content, pH, and alkalinity. In practise, the baseline requirements for each of the aforementioned metrics are set at the beginning of the regular WDS monitoring. Contamination is then indicated by considerable deviations from the base line. The study's goal was to determine whether it is feasible to develop a smart monitoring system powered by artificial intelligence (AI) that would successfully enable water operators to ensure near real-time quality control for the early detection of chemical and/or biological pollution and risk management. The study's goal was to see whether it was possible to develop an artificial intelligence (AI)-based smart monitoring system. The study's purpose was to investigate the viability of building a sophisticated monitoring system that would allow water operators to perform quality control in near real time. Artificial Neural Networks (ANN) and other sophisticated pattern recognizers like Support Vector Machines (SVMs) are examples of cutting-edge sensor technologies that have been implemented as solutions to identify anomalies and estimate the severity of such irregularities.**

Keywords: *Artificial Intelligence, SVM, Water Distribution System, Smart Monitoring System.*

I. INTRODUCTION

The constraints of traditional techniques for determining the appropriate position of water quality sensors in WDS and the restrictions of the performance that might be anticipated from a network of water quality sensors were the subject of research that was carried out [1]. The results of this research were analysed. The largest water distributor, has designated a piece of its distribution system as a Innovation Playground (IP) demonstration network for real-time water quality monitoring. With the assistance of this network, the company is aiming to boost innovation. On the IP, a network of 44 sensors has been placed, and each sensor is linked to a computer through GPRS. This allows event detection algorithms to be applied to the data that has been acquired from the sensors. It was conceived as an online sensor network, which made it practicable to pinpoint the specific site of a pollutant's propagation within the distribution network, detect it early, and react promptly [2].

When trying to identify instances of contamination, it is vital to have the capacity to differentiate between significant changes in the observed parameter values and the regular daily and/or seasonal oscillations in those values. Because of this, it is absolutely necessary to make use of certain algorithms, the majority of which are developed using statistical or stochastic methods. Artificial neural networks (ANNs), a kind of model that is data-driven and focuses on modelling and prediction, have been successfully employed in recent years for water quality assessment that attempts to improve the EDS. This is due to the fact that ANNs are data-driven and place an emphasis on modelling and prediction. After developing a model with the assistance of multivariate artificial neural networks (ANN), Bayesian theory was used to classify the occurrences [3]. Bayesian belief networks (BBNs) in conjunction with pH, conductivity, and turbidity were used for the purpose of E. coli event detection. In order to develop a technique that is applicable across the whole of the system, the binary signals that were generated as a result of the independent event detection performed by each sensor were merged with a source identification model that was founded on an algorithm for backtracking. The computer algorithm analyzed the differences

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in sensor alerts in order to determine the likelihood that an event would take place. To improve the quality of the study conducted on the subject, make use of adaptive updating dynamic thresholds. These thresholds are determined as a function of the error standard deviation of the sliding window and are dependent on the size of the sliding window [4].

The purpose of this project is to create and show the practicality of an intelligent system for monitoring water quality metrics capable of identifying biocontamination in its early stages. Additional goals include the concept of dynamic thresholds, the visualization of detected anomalies on ANNs, and considerations for the interconnected changes of various water quality parameters (such as the concentration of free chlorine, pH, alkalinity, and TOC caused by the pollutant) [5]. These are all examples of objectives that are related to the pollutant. In addition to that, the purpose of this research is to investigate the connections between the numerous shifts in water quality that have been observed.

II. RELATED WORKS

This study's objective is to simulate the spread of biocontamination risk across the Water Distribution System (WDS) by identifying the sources of the risk and modelling potential responses to it. This will be accomplished by the modelling of the propagation of bio-contamination risk inside the WDS in accordance with real environmental conditions [6]. This will be done in order to accomplish this goal. The failure to take into consideration the possibility of the bulk of the pollutants degrading chemically leads to an overestimation of the population that is exposed to them. Ignoring dynamic responses also leads to this overestimation. As a result of this, in contrast to other event detection models, the study presented here takes into account the connected change in a variety of water quality indicators that are caused by the mixing of the pollutants. These measurements include the total organic carbon content, pH, and alkalinity. Free chlorine content is also taken into consideration. In practice, the baseline requirements for each of the aforementioned metrics are set at the beginning of the regular WDS monitoring [7].

Then, deviations that are significantly different from the norm point to the possibility of contamination. The objective of the research was to determine whether or not it would be possible to develop a smart monitoring system that would make use of artificial intelligence (AI) to enable water operators to successfully ensure near real-time quality control for the early detection of chemical and/or biological pollution and the management of preventative risks [8]. The objective of the study was to determine whether or not it would be possible to develop an intelligent monitoring system that would provide water operators the ability to do quality control in a nearly real-time manner. This makes sense given that it was done in order to accomplish the same aim for which the system was first built, hence it was carried out to achieve that goal. Using Artificial Neural Networks (ANN) and other sophisticated pattern recognizers like Support Vector Machines (SVMs), modern sensor technology has been put to use to identify and classify anomalies [9].

approaches that are based on artificial intelligence have the potential to greatly reduce the costs that are connected with the development of water supply and sanitation systems. These approaches may also assist assure compliance with regulations that regulate the treatment of drinking water and wastewater [10]. Research on predicting and forecasting water quality has received a significant amount of attention recently as part of an effort to cut down on the amount of pollution that is present in the water. In this section, the innovative aspect of the approach that was suggested is analyzed in order to demonstrate how it may be used to the construction of a green environment that is not only desirable but also enduring throughout the course of time. The provision of a method that is reliable for determining the quality of drinking water is the objective of this project. In this experiment, the water quality index (WQI) was categorized using feed-forward neural networks (FFNN) and K-nearest neighbors. In addition, an adaptive neuro-fuzzy inference system (ANFIS) was used to make a prediction about the WQI [11]. After the analysis was finished, only seven of the eight significant parameters in the dataset had values that could be considered significant. This was a significant reduction from the original number. The recommended strategy was built with these statistical considerations in mind when it was developed. According to the findings of the predictions, the ANFIS model was more successful than the others in predicting WQI values. On the other hand, the FFNN approach achieved the highest degree of accuracy possible (100%) and rapidly became the standard means of determining water quality classification (WQC). In addition, the WQC classification performance of the FFNN model was more reliable than that of the ANFIS model, which properly predicted the WQI. The fact that the FFNN model performs so much better than its competitors is proof of this. The FFNN model did the best in terms of accurately forecasting WQC (100%), whereas the ANFIS model performed wonderfully throughout the tests, with a regression coefficient of 96.17%. The FFNN model was successful in predicting WQC accuracy. It has been hypothesized that this technology, which makes use of the most recent and cutting-edge AI algorithms, would have certain uses in the administration and treatment of water.

In recent years, the quality of the water has steadily declined, which may be directly attributed to the presence of a number of different contaminants. It is now very necessary to make water quality forecasts and do simulations in order to cut down on water pollution. The most cutting-edge applications of artificial intelligence (AI) were used in this work to provide accurate predictions for the water quality classification (WQC) and the water quality index (WQI). Models based on artificial neural networks have been developed in order to provide an accurate prediction of the WQI. These models make use of techniques pertaining to deep learning. the nonlinear autoregressive neural network, also known as NARNET, and the long short-term memory, also known as LSTM [12].

The forecasting of WQC uses three different types of machine learning: support vector machine (SVM), K-nearest neighbor (K-NN), and naive bayes. The developed models were evaluated based on a wide range of statistical criteria, and the dataset that was used had seven characteristics that were considered essential. The findings indicate that the models that were recommended are able to properly identify the water quality based on the enhanced robustness of the data, in addition to accurately anticipating WQI. This was determined by analyzing the outcomes of the study. When it came to forecasting WQI values, the results of the predictions showed that the LSTM model performed about slightly better than the NARNET model, with the SVM approach having the highest accuracy (97.01%). The findings also showed that the NARNET model performed approximately slightly worse than the LSTM model. The use of the NARNET paradigm acts as proof for this proposition. During the testing phase, the accuracy of both the NARNET and LSTM models was comparable, with only a slight difference in the regression coefficient (RNARNET = 96:17% and RLSTM = 94:21%). An investigation that has this much potential could provide conclusions that end up having enormous repercussions for the way water resources are managed [13].

Researchers have been hard at work developing innovative strategies and improving the accuracy of older models in order to improve their ability to produce accurate forecasts of the river's water quality. Recent occurrences have shown how successful approaches based on artificial intelligence (AI) may be in accomplishing this objective. The purpose of this comparison was to determine which of the two In the first part of this review research, an overview of each model as well as an explanation that goes along with it is included. The effectiveness of the model as a tool for simulating the water quality of rivers is then investigated in a few studies that were only recently published. The results of these investigations were very recently compiled. For the purpose of this investigation, 51 journal papers published between the years 2000 and 2016 were selected because they studied the use of artificial intelligence (AI) models to predict river water quality, either on their own or in combination with other AI models. All of these articles' analyses of predictor selection, data normalization, the separation of train and test data, modelling methodologies, evaluations of prediction time step procedures, and other aspects are taken into account [14].

The purpose of the research was to determine whether or not integrating different models would improve their capacity to make accurate forecasts of the occurrence of occurrences in the future. The freshly published works category included 31 of the 51 articles that were selected, making up about 60% of the total. This category accounted for almost all of the total. Prior to the 20th of February in 2016, the topic had been referenced in the periodicals in question a maximum of 1716 times at the most. Out of all the different modelling techniques, the ANN and WANN models were the ones that were used as single models as well as hybrid models the most often. This was true for both of these types of models. The modelling of dissolved oxygen (DO) and suspended silt in river systems gained increased attention in the research that were analysed for this article. The modelling of the water quality comprised a total of 23 separate tests, and the data that was employed included daily time periods. This research covers all 13 conceivable forms of artificial intelligence, and it includes both single and hybrid models. It consists of an in-depth review of the efficacy and reliability of several modelling strategies for replicating significant water quality data. It offers a comprehensive examination of the use of AI algorithms to the modelling of river water quality simulations. In addition to this, it investigates the possibility of using AI methods to the modelling of lake water quality. In addition to this, it creates an aquatic habitat similar to that of the river [15].

III. RESEARCH METHODOLOGY

Figure 1 depicts the approach that has been shown to be the most effective for this inquiry.

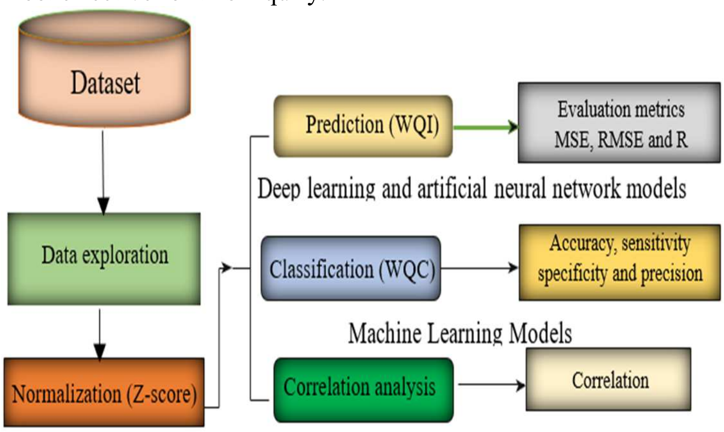

Fig. 1: Framework of the proposed methodology.

A. Dataset

The data collection that was used in this investigation was compiled using information obtained from a selection of important Indian historical sites. From 2005 to 2014, it collected a total of 1679 samples from across a number of Indian states. There are seven key parameters included in the dataset, and they are as follows: Conductivity, pH, dissolved oxygen (DO), total and faecal coliform, and nitrate are some of the parameters that are monitored and analyzed. Other parameters include coliforms. The Indian government compiled all of the relevant data in order to ensure that the drinking water adhered to all of the standards concerning its level of cleanliness. These particulars, including the numbers and the statistics, have been compiled.

B. Data Pre-processing.

The processing step is an essential component of the data analysis process since it enables an improvement in the data's overall quality. During this stage of the process, the WQI was calculated by making use of the most relevant factors from the dataset. After that had been collected. In order to get higher levels of precision via the process of data normalization, the z-score strategy has been used.

C. *Water Quality Index Calculation.*

When calculating the Water Quality Index (WQI), numerous elements that have a significant impact on the overall quality of the water are considered. In the course of this inquiry, a suggested model is evaluated against a previously published dataset using seven essential criteria pertaining to water quality. In order to calculate the WQI, the following formula was utilized: $WQI = \frac{\sum_{j=1}^{n} \times w_i}{\sum_{j=1}^{n} w_i}$

The letter N represents the number of factors considered while calculating the WQI on the quality rating scale I constructed for each parameter using the equation. This scale rates the quality of each parameter that follows is denoted by the letter qi, and the unit weight that I generated for each parameter by using the equation is denoted by the letter we.

$$q_i = 100 \times \left(\frac{V_i - V_{ideal}}{S_i - V_{ideal}}\right)$$

Where: The measured value of the metric i, which was denoted by the symbol Vi, was found in the water samples that were examined. Table 1 reveals the suggested standard value of the parameter Si, and pure water reveals the ideal value of the parameter Ideal. Si is the proposed standard value, and Ideal is the ideal value. The value that was measured for the parameter i in each of the water samples that were analyzed is referred to as vi. When the water is transparent, the parameter i ought to have the value V assigned to it.

Table 1: Permissible limits of the parameters used in calculating WQI

Parameters	Permissible limits
Dissolved oxygen, mg/l	11
pH	8.9
Conductivity, μS/cm	1000
Biological oxygen demand, mg/l	6
Nitrate, mg/l	51
Fecal coliform, Cfu/100 ml	101
Total coliform, Cfu/100 ml	1100

D. *Artificial Neural Network (ANN) Model.*

In a wide range of engineering applications, neural network (NN) models are often used as especially effective machine learning approaches for time-series prediction. This is because NN models are based on the concept of neural networks. This is due to the fact that the concept of neural networks acts as the basis for NN models. This is because the fundamental underpinning of NN models is the concept of neural networks. The three independent layers that comprise the ANN model are an input layer, one or more hidden layers, and an output layer. These strata are ascending in order.

To exercise control over the activity of the neurons that may be located in each buried layer, weight and bias parameters must be used. In this approach, the activation function may be used to transmit data from the hidden layer to the output layer. Learning processes are used to choose the weights of the NN structure. The mean square error (also known as MSE) is one of the performance measures that is considered throughout the process of selecting the weights.

Table 2: Parameter unit weights.

Parameters	Unit Weight (w_i)
Dissolved oxygen	0.2216
pH	0.2605
Conductivity	0.0026
Biological oxygen demand	0.04429
Nitrate	0.0498
Fecal coliform	0.0228
Total coliform	0.0026

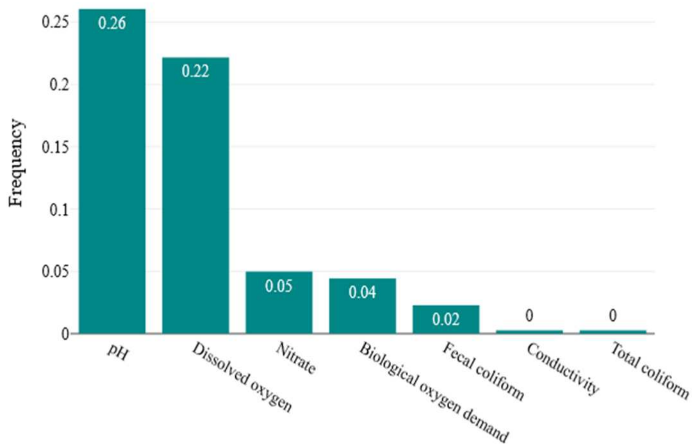

Fig. 2: Parameter unit weights.

The NARNET model, which is a multilayer feedforward network, has recently gained a great deal of popularity. It starts with a preliminary estimate of the weight, which is then modified based on the most current data. This has the direct impact of introducing an additional component of uncertainty into the process of prediction that the NN model employs. The procedure begins with the network being trained several times using a variety of initially produced configurations that are chosen at random, and it is then followed by the average of the outputs of these training sessions. Before work can begin on the NARNET model, it is necessary to ascertain the total quantity of hidden layers and nodes. Figure 2 is a depiction of the NARNET model approach, which includes a greater number of inputs than only the four hidden layers that are often indicated for the vast majority of research datasets. This technique may be found in the NARNET model. A description of the NARNET time series model is included inside the equation.

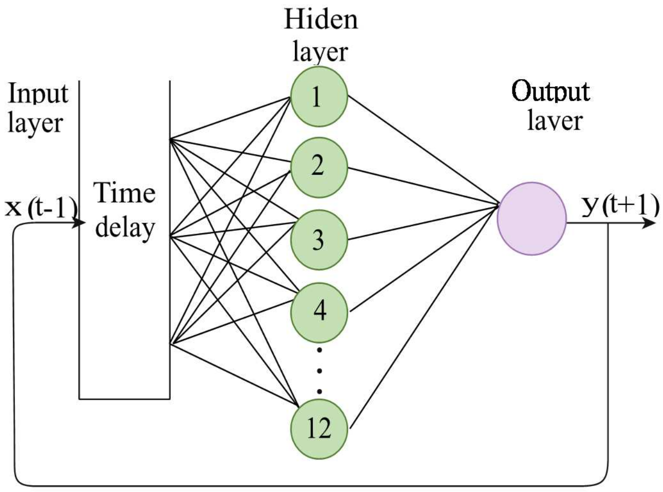

Fig. 3: Computation of the NARNET model

In order to provide forecasts about the WQI, the NARNET model was developed by the use of this data. Since the WQI seems to have characteristics that are time series, it is recommended to use the NARNET model to make predictions about the WQI. Table 2 provides you with access to the most important model parameters for your perusal.

IV. RESULT ANALYSIS AND DISCUSSION

The value 16.85 is the value of the positive coefficient for the variable Unit Weight (wi). As a result of this, it seems that the chance that "Dissolved oxygen" is the dependent variable rises as the Unit Weight (wi) increases. This impact does not seem to exist, according to the p-value of.283, which is despite the fact that it is not statistically significant. According to the odds ratio of 20893523.86, the likelihood that the dependent variable "Dissolved oxygen" would increase by 20893523.86 times for every unit increase in the variable Unit Weight (wi) will increase by 20893523.86 times for every unit. Table 3 and 4 gives the data parameters based on Classification and Logical Regression respectively.

Table 3: Classification Table

		Predicted		
		Not Dissolved oxygen	Dissolved oxygen	Correct
Observed	Not Dissolved oxygen	5	1	83.33 %
	Dissolved oxygen	1	0	0 %
	Total			71.43 %

In order to make a prediction of the value "Dissolved oxygen," logistic regression analysis was used to study the influence of Unit Weight (wi) on variable Parameters. This was done in order to anticipate the value. According to the findings of a logistic regression analysis (Chi2(1) = 2.02, p.155, n = 7), the model as a whole does not meet the criteria for statistical significance.

Table 4: Logistic Regression

	Coefficient B	Standard error	z	p	Odds Ratio	95% conf. interval
Unit Weight (wi)	16.85	15.71	1.07	.283	20893523.86	0 - 488278819073154700000
Constant	-4.28	3.46	1.24	.216		

A log-rank test was used to examine the distributions of time until an event took place for each of the following groups: dissolved oxygen, pH, conductivity, biological oxygen demand, nitrate, total coliform, and faecal coliform. According to the results of the Positive if greater than or equal log-rank test carried out on the present data, which produced a p-value of 1, the distribution of the amount of time remaining before the occurrence of the event does not vary substantially between the groups. As a direct result of this, there is no evidence to suggest that the null hypothesis is inaccurate.

V. CONCLUSION

It is essential for the conservation of the environment to have accurate models and forecasts of the water quality in Water Distribution System, both of which may be accomplished via the deployment of AI algorithms. Data was collected from numerous areas around the country in order to construct artificial intelligence models. These models were developed with the intention of anticipating and categorizing probable drinking water quality in rivers in various Indian states. The WDS was used to identify seven critical parameters, including DO, pH, conductivity, biological oxygen demand, nitrate, and total and faecal coliform. DO was one of the metrics that was determined. These components were thought to be significant indications of the water's overall quality at the time. There is a good chance that the use of AI and ANFIS algorithms of a higher level of sophistication will be of assistance in the creation of new methods that may help to the formation of a more secure setting. This strategy, which was suggested by ANFIS, included the use of algorithms that became progressively more complicated as they attempted to anticipate WQI. Utilizing the FFNN method resulted in accurate classification of the WQC data. The strategy that was offered was scrutinized in great detail and evaluated using statistics.

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